



Solve with us 6.2 - Not Graded



This assignment will not be graded and is only for practice.

Key Points: 1Key Points: 1

- Let $T: V \rightarrow W$ be a linear transformation. We define kernel of T (denoted by $\ker(T)$) and image of T (denoted by $\text{Im}(T)$) as follows:

$$\text{Ker}(T) = \{v \in V \mid T(v) = 0\}$$

$$\text{Im}(T) = \{w \in W \mid \text{there exists } v \in V \text{ for which } T(v) = w\}$$

$$\{w \in W \mid \text{there exists } v \in V \text{ for which } T(v) = w\}$$

- T is injective (i.e., T is a monomorphism) if and only if $\text{Ker}(T) = \{0\}$.
- A linear transformation $T: V \rightarrow W$ is surjective (i.e., T is epimorphism) if and only if $\text{Im}(T) = W$.
- $\text{Nullity}(T) = \dim(\text{Ker}(T))$ and $\text{Rank}(T) = \dim(\text{Im}(T))$.
- Let $T: V \rightarrow W$ be a linear transformation. Then

$$\text{Rank}(T) + \text{Nullity}(T) = \dim(V).$$

This is known as the rank nullity theorem.

- Let $T: V \rightarrow W$ be a linear transformation. T is injective if and only if $\text{Nullity}(T) = 0$. T is surjective if and only if $\text{Rank}(T) = \dim(W)$.

Consider the following linear transformation:

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2 \quad T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$T(x, y, z) = (2x + 3z, 4y + z)$$

$$T(x, y, z) = (2x + 3z, 4y + z)$$

Answer the following questions:

1 point

Which of the following Options is true?

[Hint: $T(x, y, z) = (2x + 3z, 4y + z) = (0, 0)$ i.e., $2x + 3z = 0$ and $4y + z = 0$]



$$\ker(T) = \{(3y, y, -4y) \mid y \in \mathbb{R}\}$$



$$\ker(T) = \{(6y, y, 4y) \mid y \in \mathbb{R}\}$$



$$\ker(T) = \{(6y, y, -4y) \mid y \in \mathbb{R}\}$$



$$\ker(T) = \{(3y, y, 4y) \mid y \in \mathbb{R}\}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\ker(T) = \{(6y, y, -4y) \mid y \in \mathbb{R}\}$$

1 point

Which of the following is a basis of $\text{Ker}(T)$?



$\{(3, 1, -4)\}\{(3, 1, -4)\}$

☐

$\{(6, 0, -4), (0, 1, -4)\}\{(6, 0, -4), (0, 1, -4)\}$

☐

$\{(6, 1, -4)\}\{(6, 1, -4)\}$

☐

$\{(3, 0, -2), (0, 1, -4)\}\{(3, 0, -2), (0, 1, -4)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(6, 1, -4)\}\{(6, 1, -4)\}$

Find the nullity of TT .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Find the rank of TT .

[Hint: use rank nullity theorem.]

No, the answer is incorrect.

Score: 0

Feedback:

– Here $\dim(V) = 3$ as $V = \mathbb{R}^3$.

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Which of the following is true?

☐

TT is both injective and surjective.

☐

TT is injective but not surjective.

☐

TT is surjective but not injective.

☐

TT is neither injective nor surjective.

No, the answer is incorrect.

Score: 0

Accepted Answers:

TT is surjective but not injective.

1 point

Which option represents the kernel and image of the following linear transformation?

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \quad T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x, 0) \quad T(x, y) = (x, 0)$$

☐

$\ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}$ $\ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}.$

○

$$\ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(0, 1)\} \quad \ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(0, 1)\}.$$

○

$$\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\} \quad \ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}.$$

○

$$\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(0, 1)\} \quad \ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(0, 1)\}.$$

No, the answer is incorrect.

Score: 0

Feedback:

• $\text{Ker}(T) = \{(0, y) \mid y \in \mathbb{R}\}.$

Accepted Answers:

$$\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\} \quad \ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}.$$

Key Points: 2 Key Points: 2

- Let $T: V \rightarrow W$ be a linear. Follow the steps below to find bases for the null space and the range space of T .

Step 1: Find the matrix A corresponding to T with respect to some standard ordered bases

$\beta = \{v_1, v_2, \dots, v_n\}$ and $\gamma = \{w_1, w_2, \dots, w_m\}$ for V and W respectively.

Step 2: Use row reduction on A to obtain the matrix R which is in reduced row echelon form.

Step 3: The basis of the solution space of $Rx = 0$ is the basis of null space of matrix A and can be obtained by finding the pivot and non-pivot columns (dependent and independent variables) as seen earlier.

Step 4: The vectors $\begin{bmatrix} c_{11} \\ c_{12} \\ \vdots \\ c_{1n} \end{bmatrix}, \begin{bmatrix} c_{21} \\ c_{22} \\ \vdots \\ c_{2n} \end{bmatrix}, \dots, \begin{bmatrix} c_{k1} \\ c_{k2} \\ \vdots \\ c_{kn} \end{bmatrix}$ form a

basis of the null space of A precisely when the vectors $v'_1, v'_2, \dots, v'_k \in \ker(T)$ where $v'_i = \sum_{j=1}^n c_{ij} v_j$ form a basis for $\ker(T)$. Use the basis obtained in step 3 to thus get a basis for $\ker(T)$.

Step 5: Recall that if $i_1, i_2, \dots, i_r$ are the columns of R containing the pivot elements, then the same columns of A form a basis for the column space of A .

Step 6: The vectors $\begin{bmatrix} d_{11} \\ d_{12} \\ \vdots \\ d_{1m} \end{bmatrix}, \begin{bmatrix} d_{21} \\ d_{22} \\ \vdots \\ d_{2m} \end{bmatrix}, \dots, \begin{bmatrix} d_{r1} \\ d_{r2} \\ \vdots \\ d_{rm} \end{bmatrix}$

form a basis of the column space of AA precisely when the vectors $w'_1, w'_2, \dots, w'_r \in \text{im}(T)$, $w'_1, w'_2, \dots, w'_r \in \text{im}(T)$ where $w'_i = \sum_{j=1}^m d_{ij} w_j$, $w'_i = \sum_{j=1}^m d_{ij} w_j$, form a basis for $\text{im}(T)$. Use the basis obtained in step 5 to thus get a basis for $\text{im}(T)$.

Consider the following linear transformation:

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3 \quad T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (x - z, 2x + 3y + z, 3y + 3z) \quad T(x, y, z) =$$

$$(x - z, 2x + 3y + z, 3y + 3z)$$

Answer the following questions 7-9 using the information given above.

1 point

Which of the matrices corresponds the given linear transformation T with respect to the standard ordered basis of $\mathbb{R}^3$ for both the domain and the codomain?

☐

$$\begin{bmatrix} 1 & 20 & 0 \\ 0 & 3 & 3 \\ -1 & 1 & 3 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & 23 & 0 \\ -1 & 3 & 3 \\ 0 & 10 & 0 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & -10 & 0 \\ 2 & 3 & 1 \\ 3 & 3 & 0 \end{bmatrix}$$

☐

$$\begin{bmatrix} 10 & -1 & 0 \\ 23 & 1 & 1 \\ 0 & 3 & 3 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 10 & -1 & 0 \\ 23 & 1 & 1 \\ 0 & 3 & 3 \end{bmatrix}$$

1 point

What will be the kernel of T ?

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$$\{(x, 0, z), (0, y, z) \mid x, y, z \in \mathbb{R}\} \{(x, 0, z), (0, y, z) \mid x, y, z \in \mathbb{R}\}.$$

☐

$$\{(z, 0, z), (0, -z, z) \mid z \in \mathbb{R}\} \{(z, 0, z), (0, -z, z) \mid z \in \mathbb{R}\}.$$

☐

$$\{(z, -z, z) \mid z \in \mathbb{R}\} \{(z, -z, z) \mid z \in \mathbb{R}\}.$$

☐
 $\{(x, 0, z), (0, -x, z) \mid x, z \in R\} \{(x, 0, z), (0, -x, z) \mid x, z \in R\}.$

No, the answer is incorrect.

Score: 0

Accepted Answers:

 $\{(z, -z, z) \mid z \in R\} \{(z, -z, z) \mid z \in R\}.$

1 point

What will be the basis of the image space of TT found by the algorithm mentioned in the Key Points ?

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 $\{(1, 0, -1), (2, 3, 1)\} \{(1, 0, -1), (2, 3, 1)\}$
☐
 $\{(1, 0, 0), (0, 1, 0)\} \{(1, 0, 0), (0, 1, 0)\}$
☐
 $\{(1, 2, 0), (0, 3, 3)\} \{(1, 2, 0), (0, 3, 3)\}$
☐
 $\{(-1, 1, 3)\} \{(-1, 1, 3)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

 $\{(1, 2, 0), (0, 3, 3)\} \{(1, 2, 0), (0, 3, 3)\}$

Check Answers

Your score is: 0/9